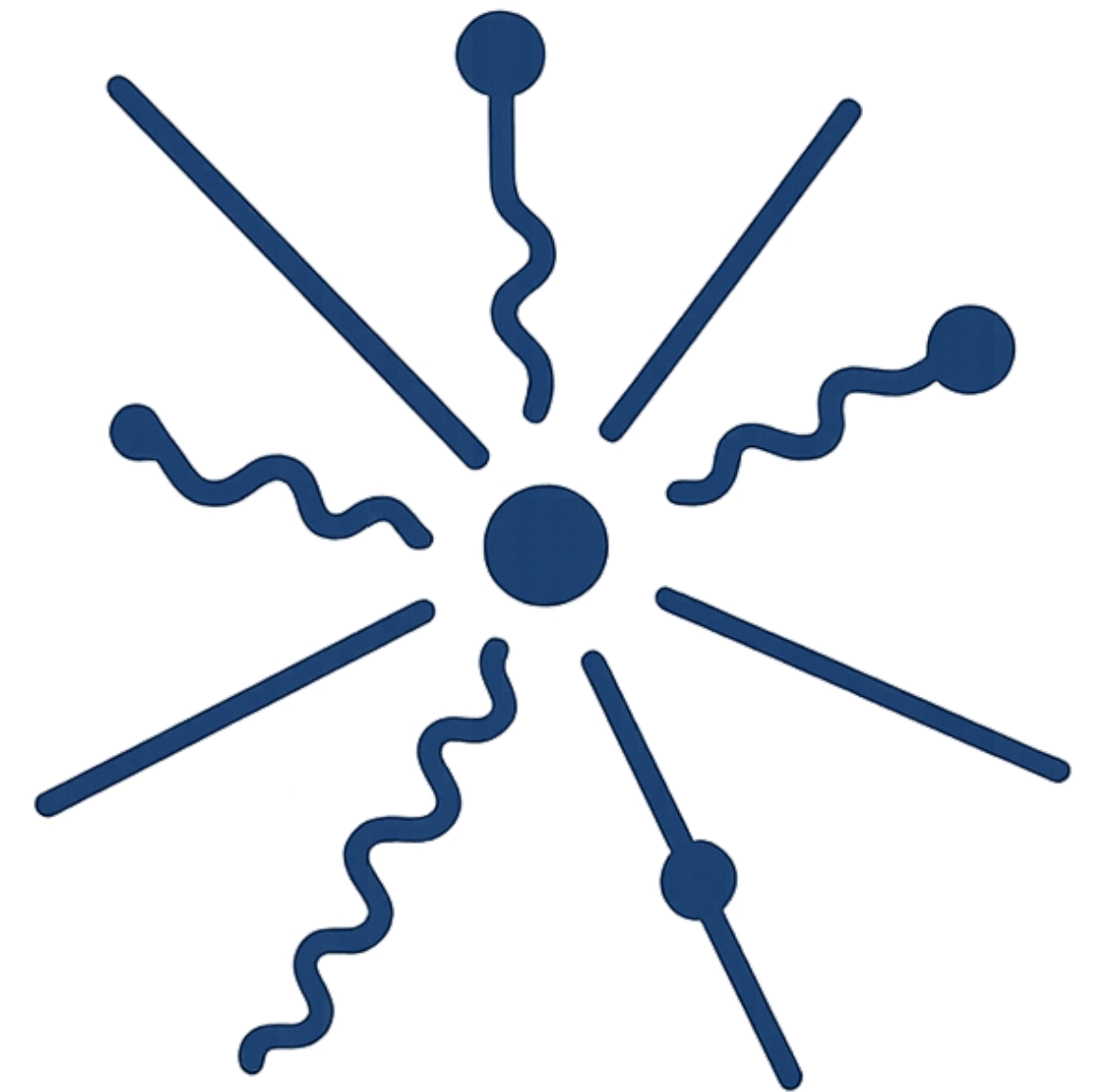
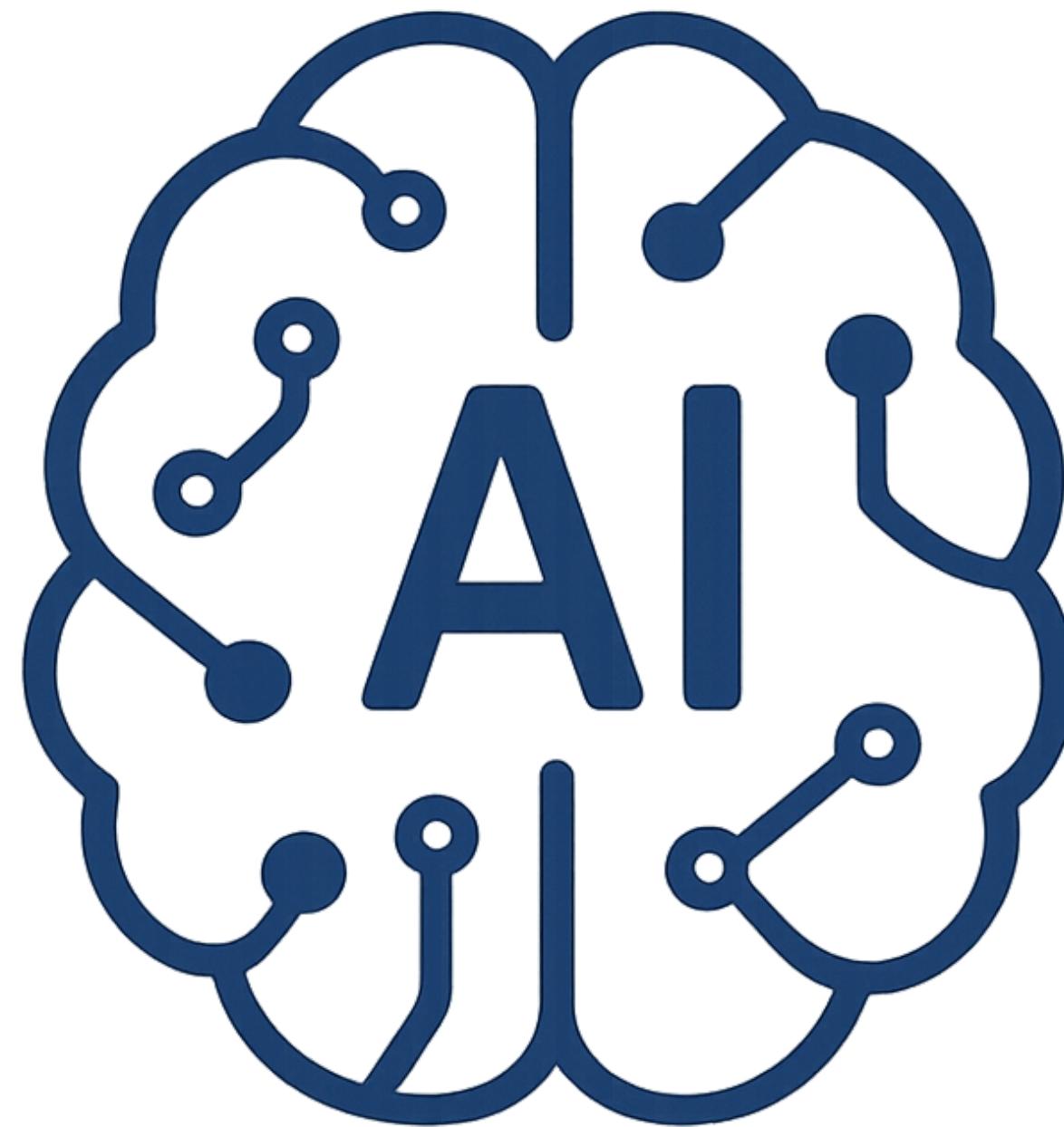
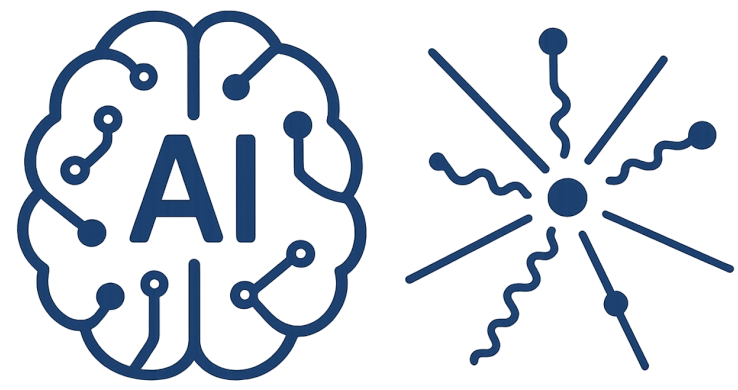


Introduction to

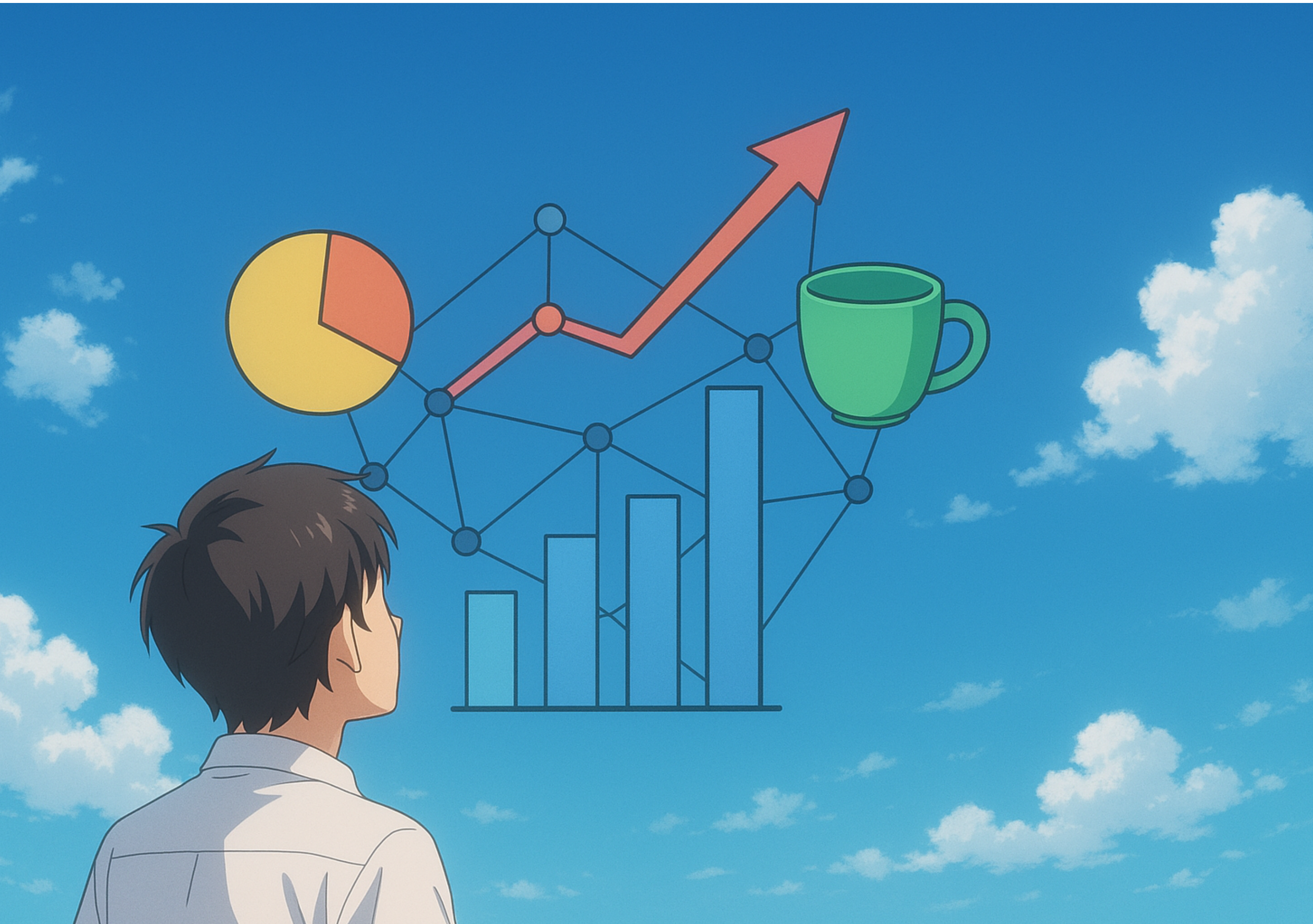
AI-Driven HEP

S. A. Fard - School of Physics (IPM)



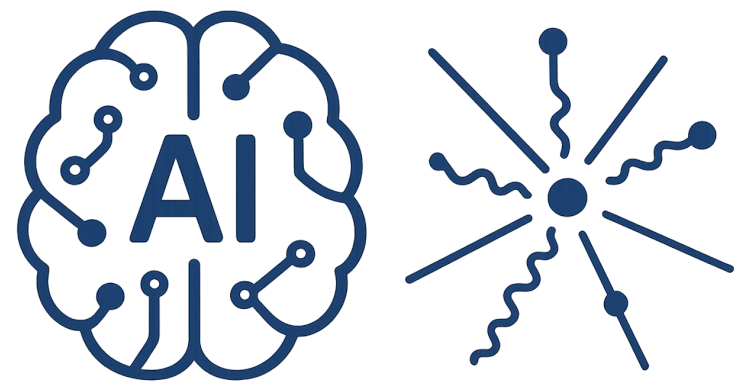


AI-Driven HEP 6



Session 6

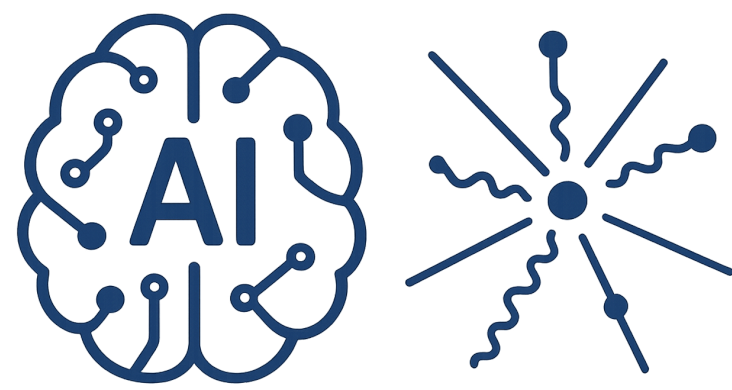
Statistical Learning



AI-Driven HEP 6

Reference

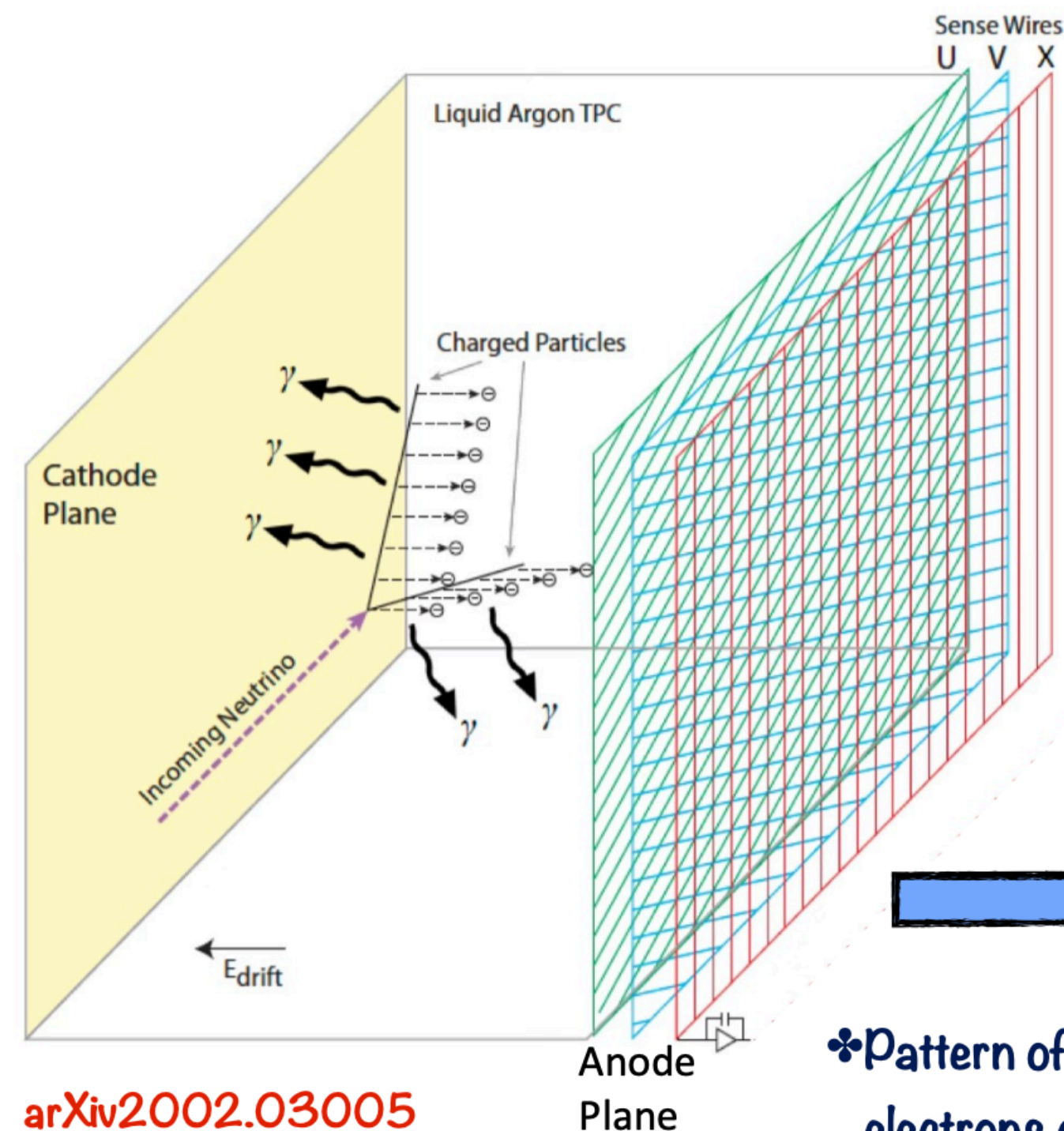
- [1] James, Gareth, et al. "Statistical learning." An introduction to statistical learning: With applications in Python. (2023)
- [2] Hastie, Trevor. "The elements of statistical learning: data mining, inference, and prediction." (2009)
- [3] Bishop, Christopher M., Pattern recognition and machine learning. springer, (2006)
- [4] Chollet, Francois. Deep learning with Python. (2021)
- [5] Shalev-Shwartz, Shai, and Shai Ben-David. Understanding machine learning: From theory to algorithms. (2014)
- [6] Tingjun Yang, A Brief Description of Data Products, ProtoDUNE Analysis Workshop, Jan 27, 2019, CERN
- [7] Gabriela Vitti Stenico, Development of the computing framework for monitoring the quality of ProtoDUNE off-line data, DUNE collaboration meeting (2025)



AI-Driven HEP 6

So far

Principle of operation of a LArTPCs - HD

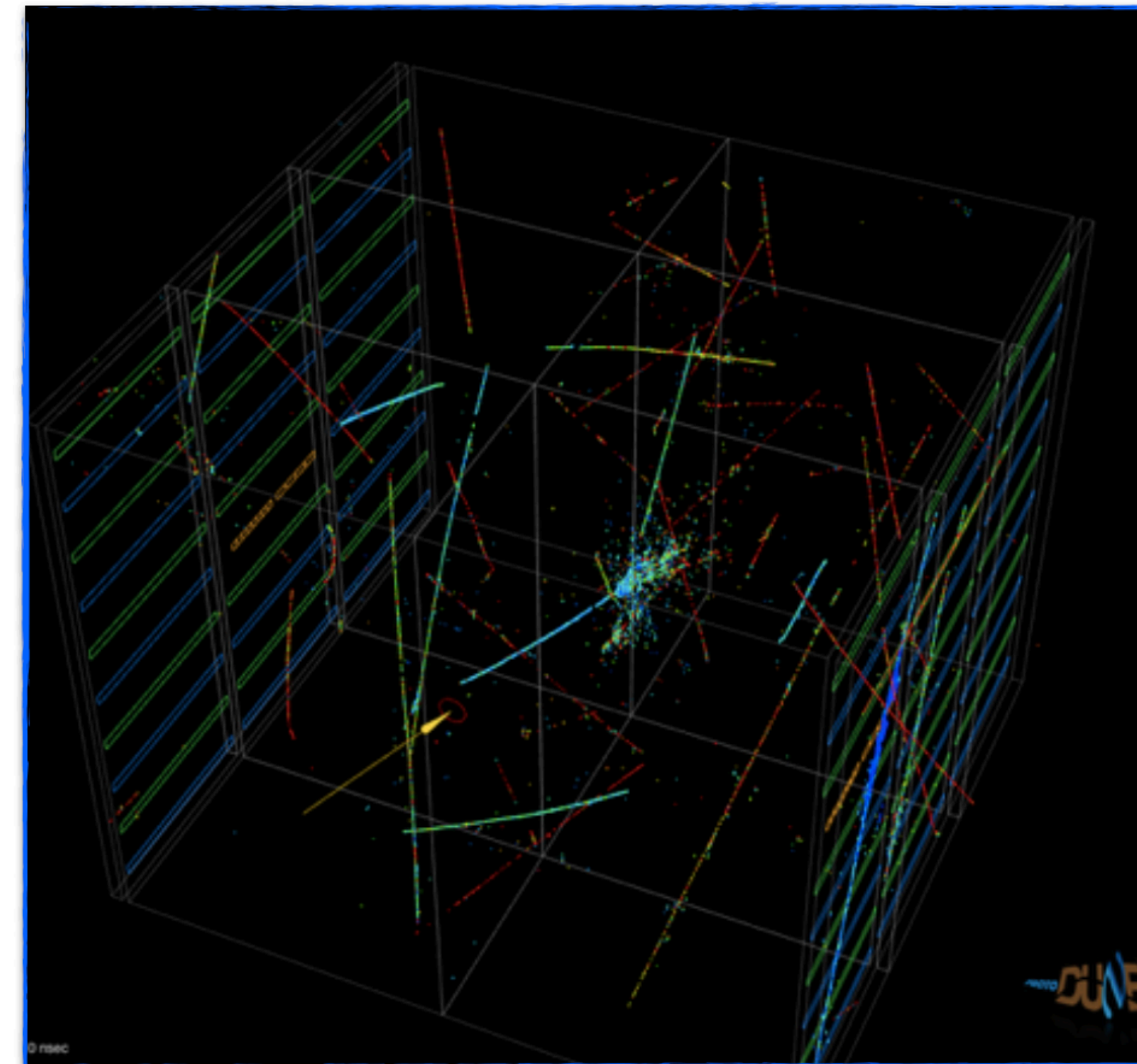


[arXiv2002.03005](https://arxiv.org/abs/2002.03005)

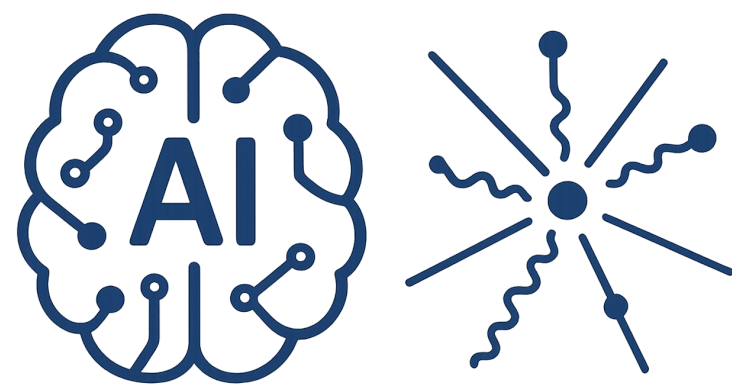
- ❖ Charged particles ionize Liquid Argon atom as they cross the detector;
- ❖ Uniform electric field drifts the ionization electrons towards the Anode Plane;
- ❖ Electronic cloud induces current in U and V wire planes (induction);
- ❖ Electrons are collected in the X wire plane (collection);

❖ Pattern of ionization and time of arrival of the electrons on the wires provide the 3D reconstruction of the particle trajectories!

Photo: DUNE Collaboration



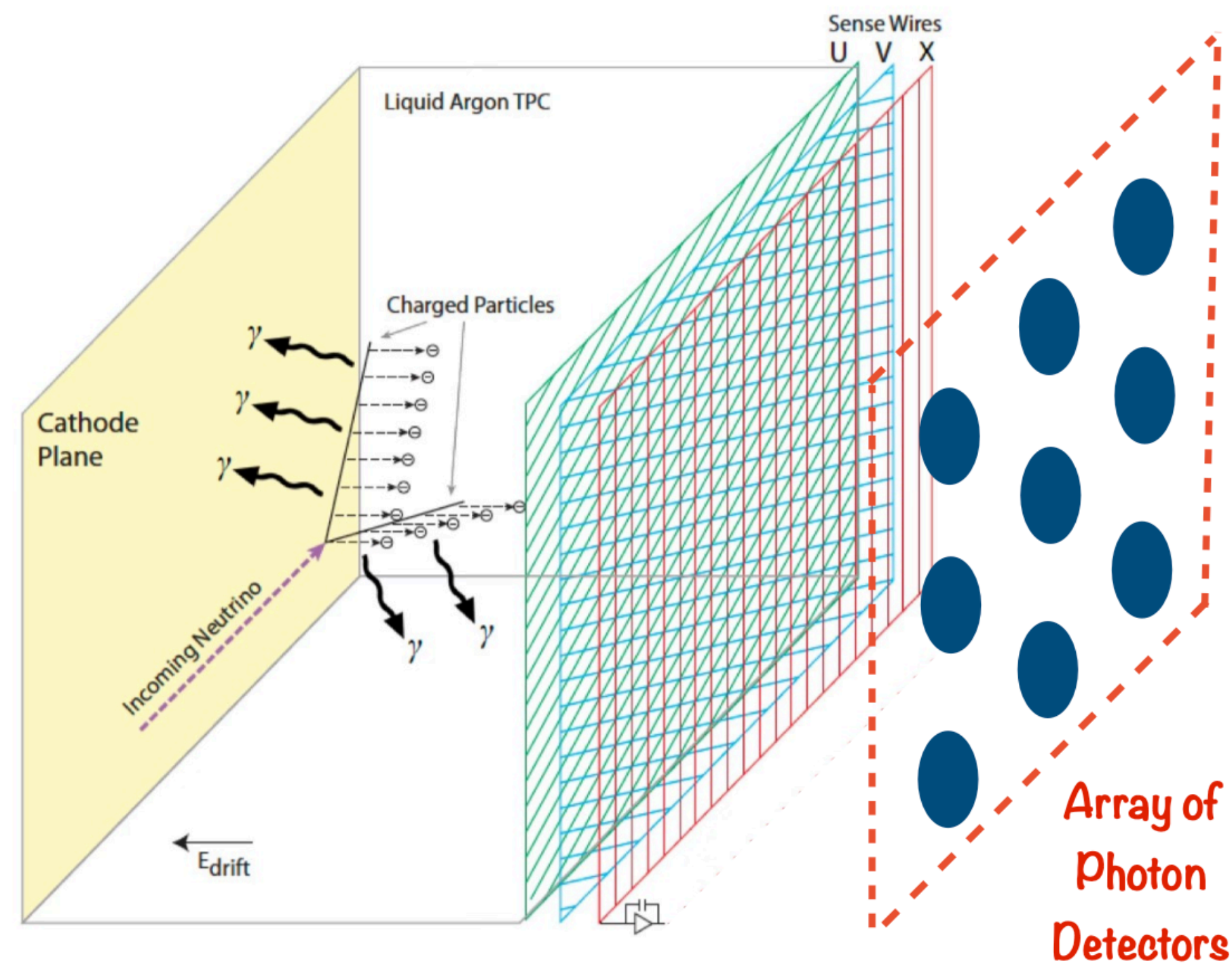
Reconstructed beam charged particle and cosmic muons interacting inside ProtoDUNE



AI-Driven HEP 6

So far

Principle of operation of a LArTPCs - HD



arXiv2002.03005

[7]

- ❖ In the process of de-excitation and recombination of the Argon atoms, scintillation light is emitted in the Vacuum Ultra Violet band;
- ❖ Light propagates through Liquid Argon and it is detected by the Photon Detection System!
- ❖ As light is faster than charge, it provides the reference time for triggering particle interaction in the detector;
- ❖ Amount of light produced is important for particle calorimetry.

ProtoDUNE's Photon Detection System is based on X-ARAPUCA technology:

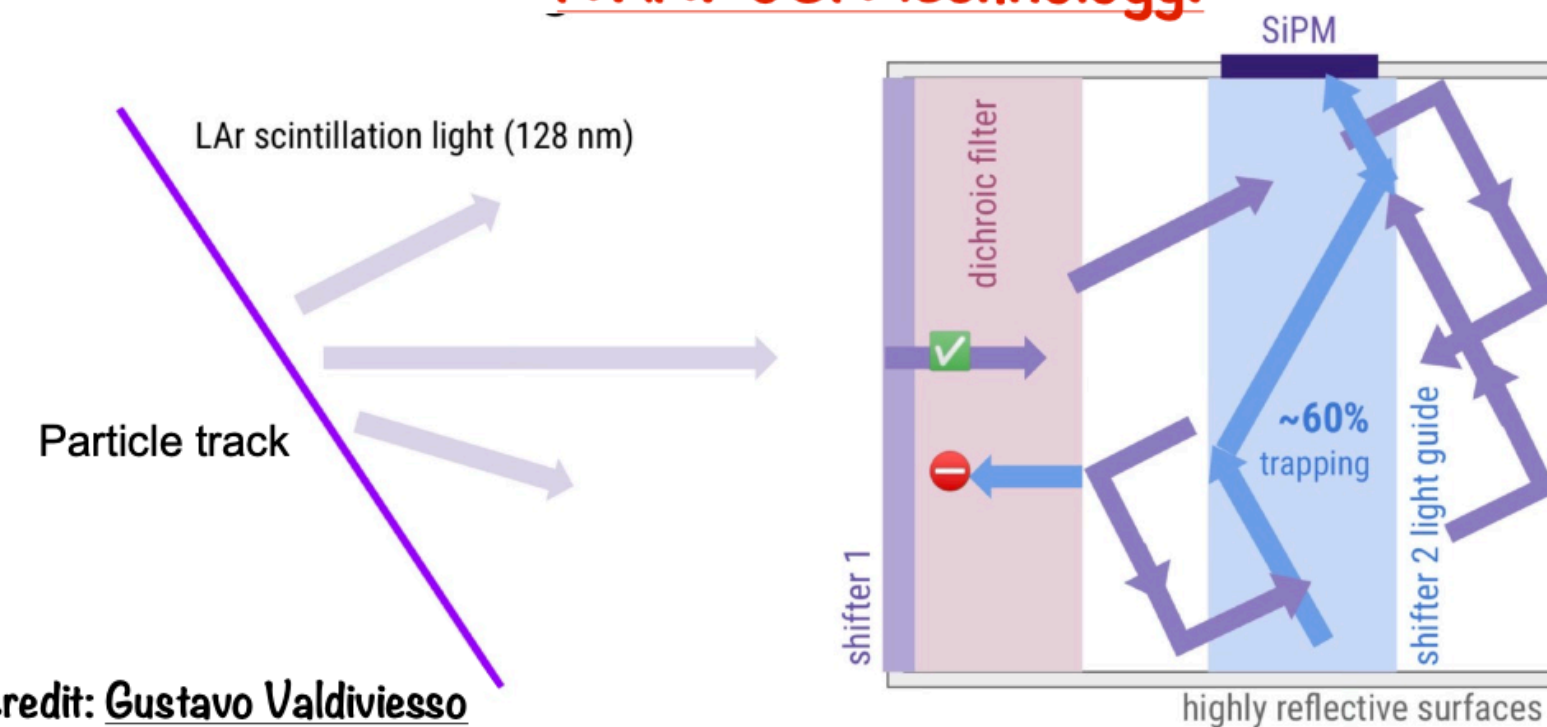
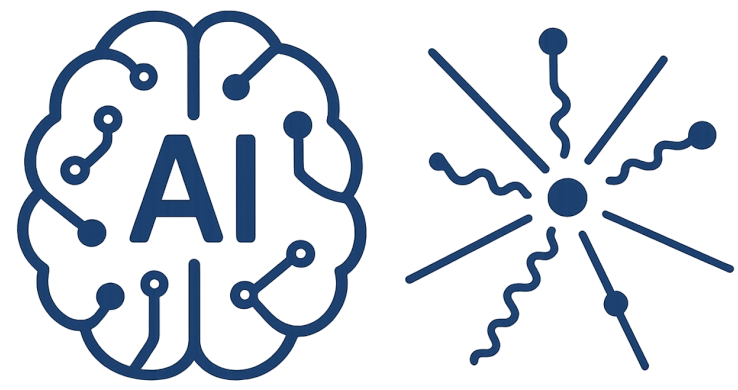
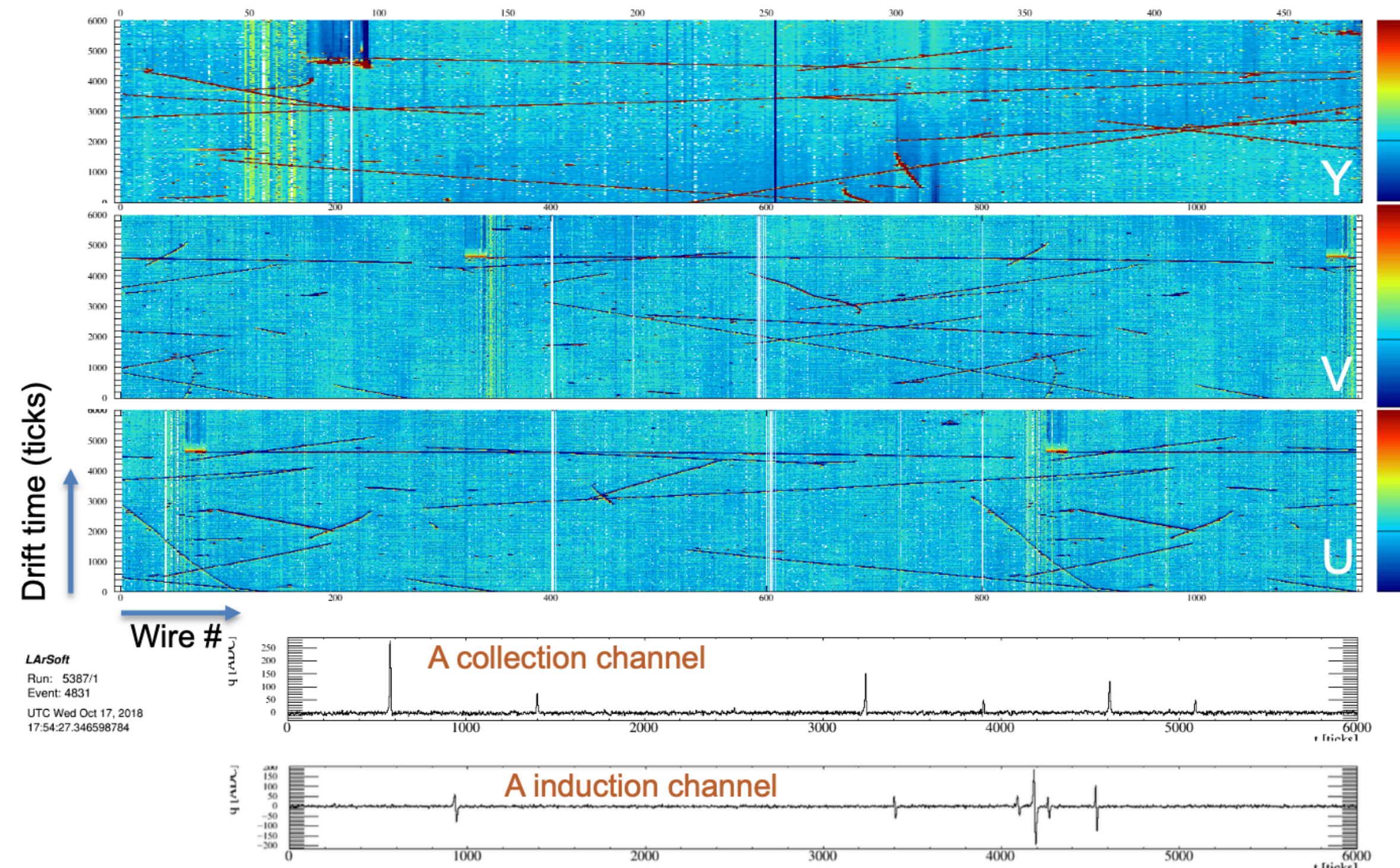


Image credit: Gustavo Valdivieso

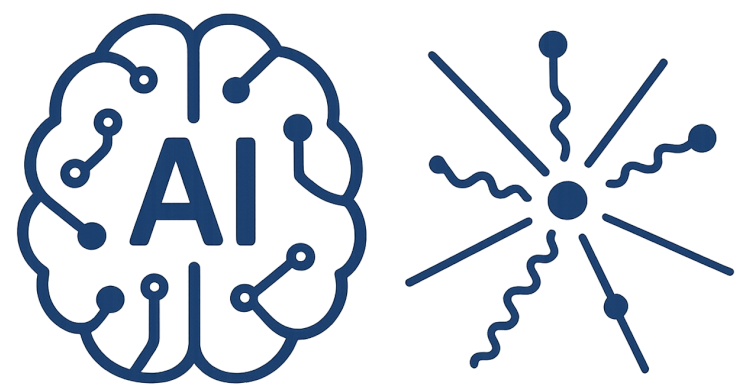


AI-Driven HEP 6

So far **raw::Digit**

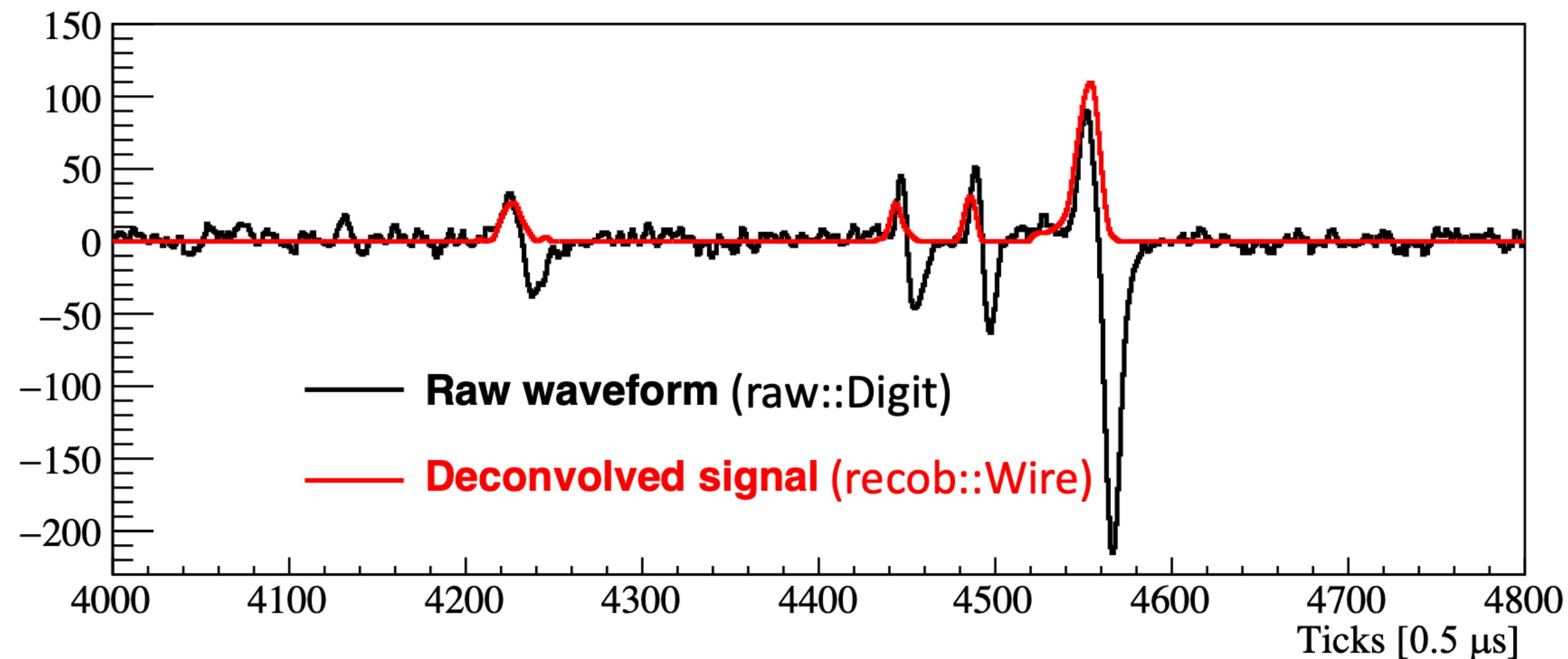


- One raw::Digit per channel, 15360 in total.
 - A raw::Digit has a vector of raw ADC counts. The size is determined by readout window, 6k by default, some runs were taken with 15k.
 - Each tick is $0.5 \mu\text{s}$.
- [6]

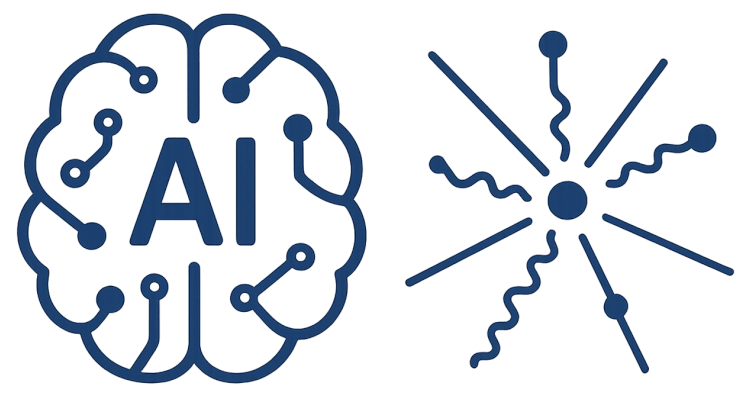


So far

recob::Wire

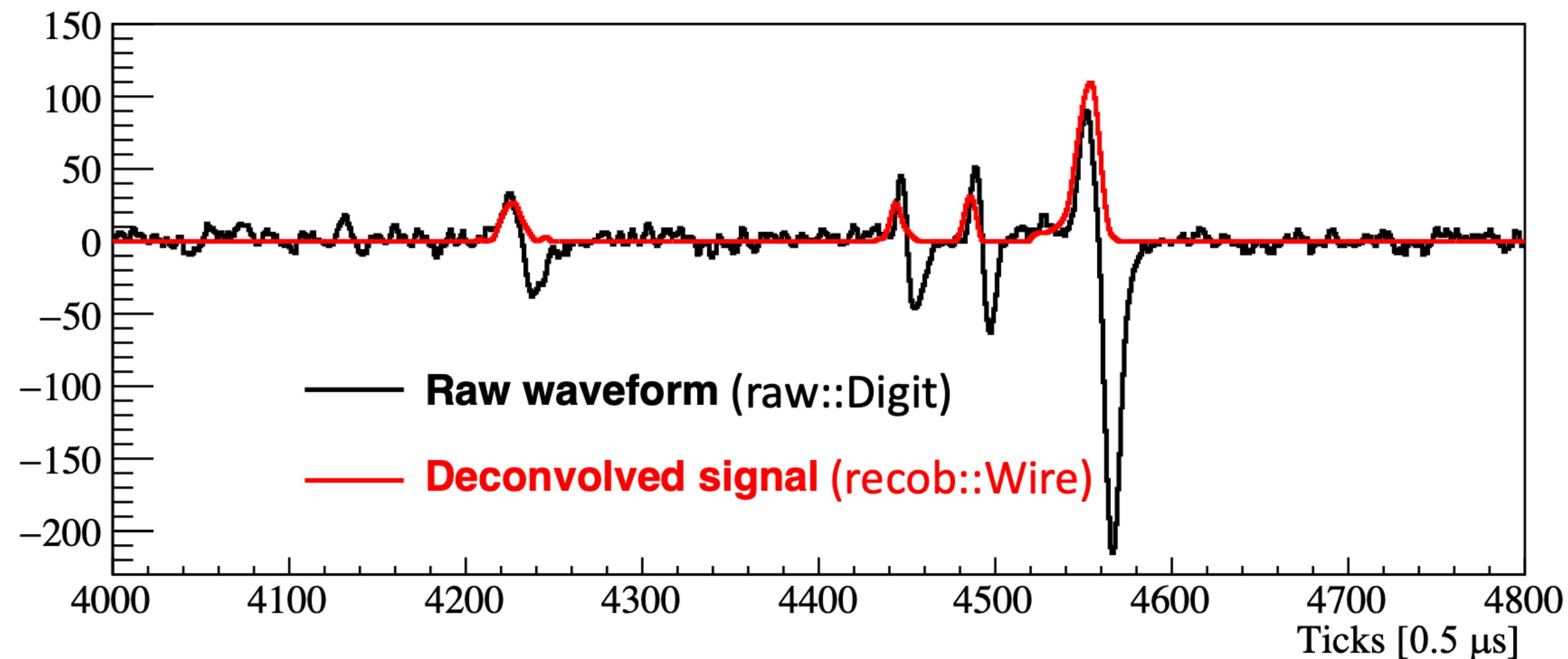


- Output of signal processing after noise filtering and deconvolution and ROI finding.
- A `recob::Wire` has a vector of float point numbers, which are deconvolved ADC counts.
- 2D deconvolution to account for induced charge on neighboring wires.

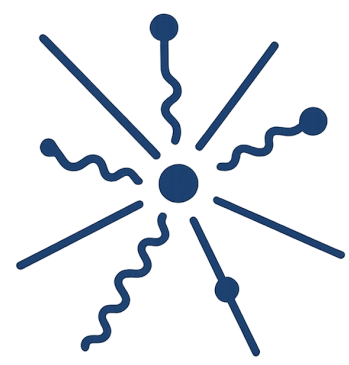
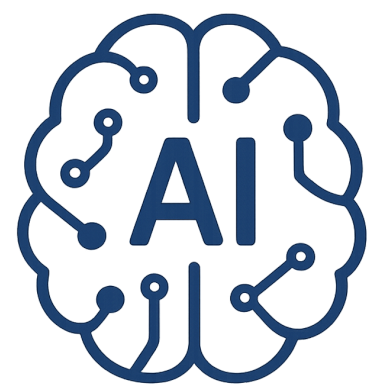


So far

recob::Wire

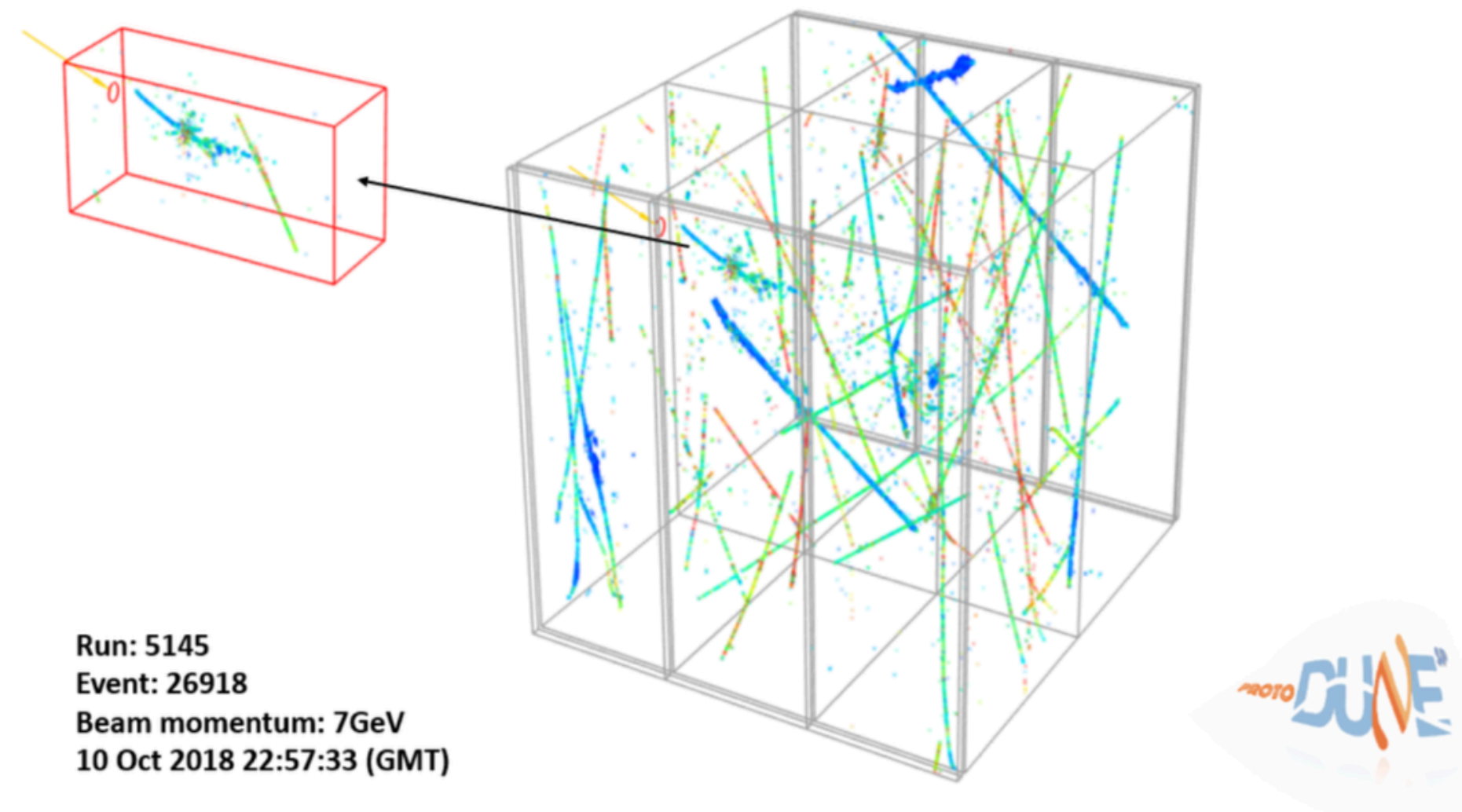


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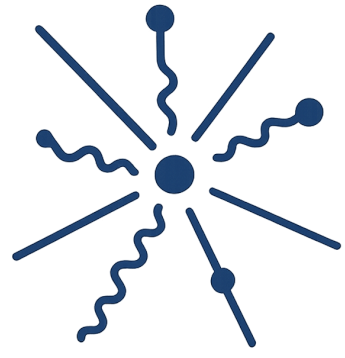
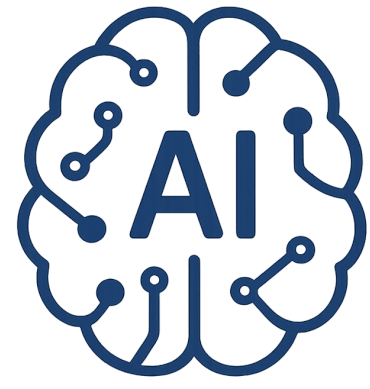


AI-Driven HEP 6

So far **recob::SpacePoint**

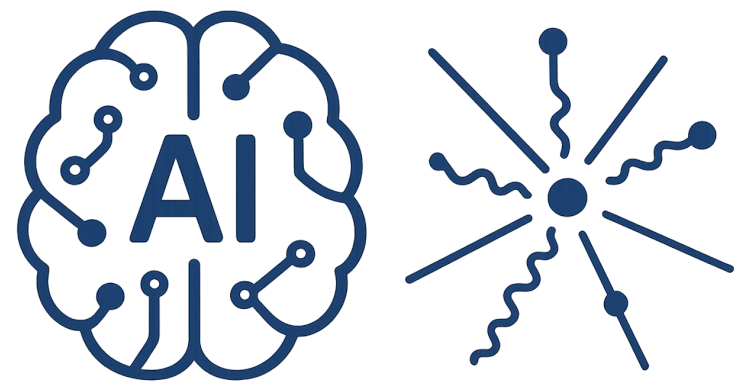


- A object to save 3D points.
- It saves the x,y,z coordinates as well as charge information.
- Used in ProtoDUNE DQM event display.
- Can be by-product of track fitting.
- SpacePointSolver, wire-cell and Pandora can make space points using hits.
- Can have associations with hits on 3 planes – help disambiguation and cluster matching.



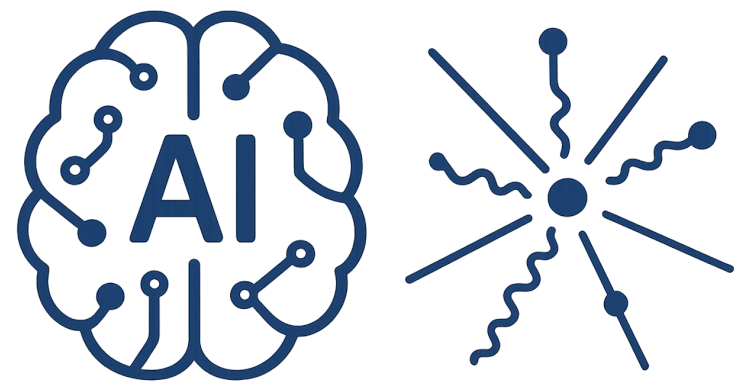
AI-Driven HEP 6

Take a deep breath



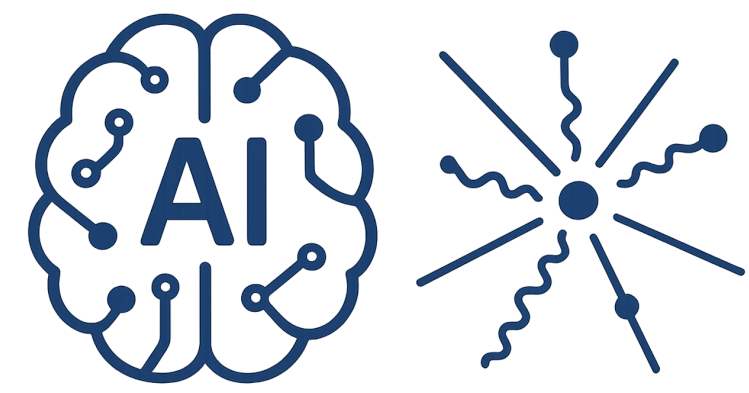
AI-Driven HEP 6

Forget about
everything you heard
so far



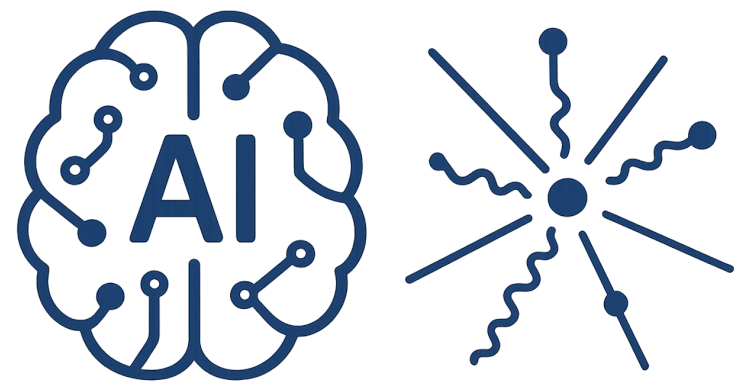
AI-Driven HEP 6

You are going to be
a data scientist
from now on



AI-Driven HEP 6

What is Statistical Learning ?



AI-Driven HEP 6

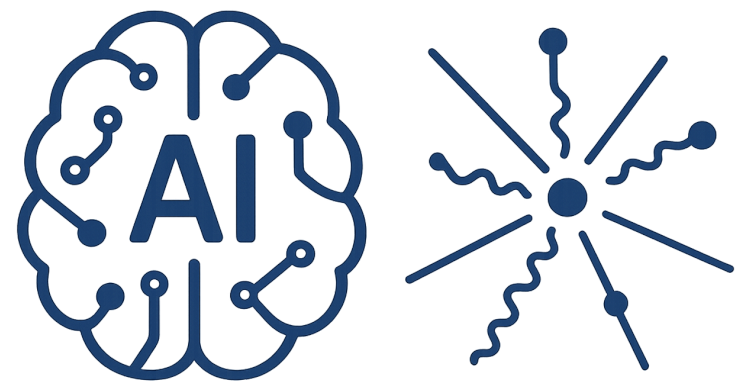
What is Statistical Learning ?

Consider

X (Input)

Y (Output)

ϵ (Uncertainty)



What is Statistical Learning ?

Consider

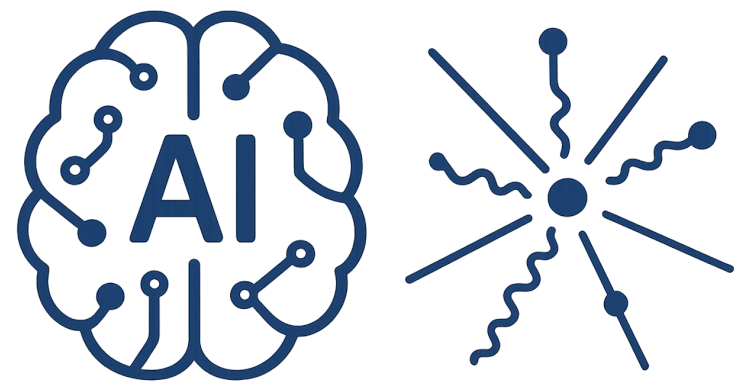
X (Input)

predictors,
independent
variables,
features

Y (Output)

response,
dependent
variable,
target

ϵ (Uncertainty)



AI-Driven HEP 6

What is Statistical Learning ?

Consider

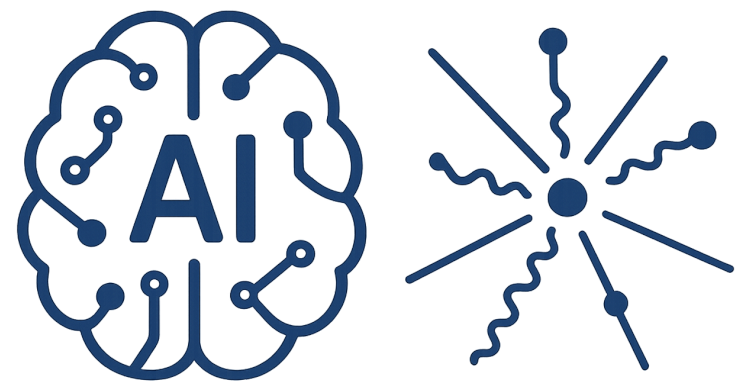
X (Input)

Y (Output)

ϵ (Uncertainty)

Statistical learning refers to a set of approaches for estimating f

$$Y = f(X) + \epsilon.$$



What is Statistical Learning ?

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X (Input)

Y (Output)

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Statistical learning refers to a set of approaches for estimating f

$$Y = f(X) + \epsilon.$$

Input

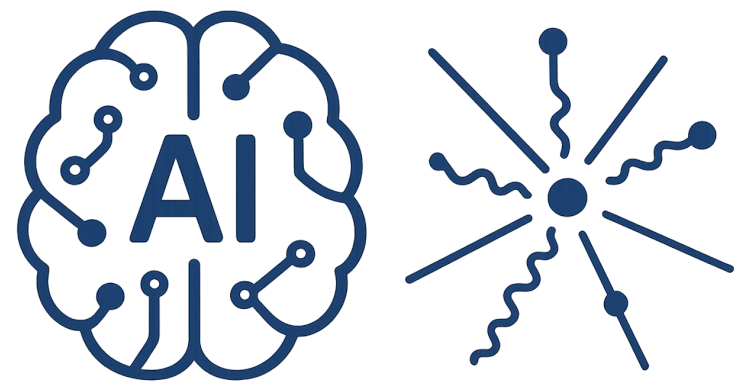
Output

Task

Architecture

Supervision

Optimization



What is Statistical Learning ?

Consider

X (Input)

Y (Output)

ϵ (Uncertainty)

Statistical learning refers to a set of approaches for estimating f

$$Y = f(X) + \epsilon.$$

Input

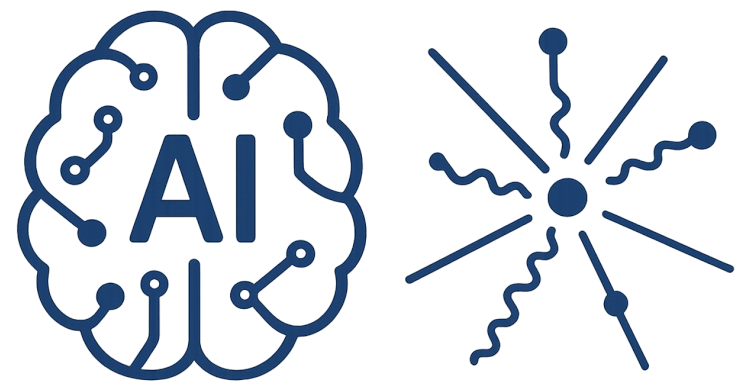
Output

Task

Architecture

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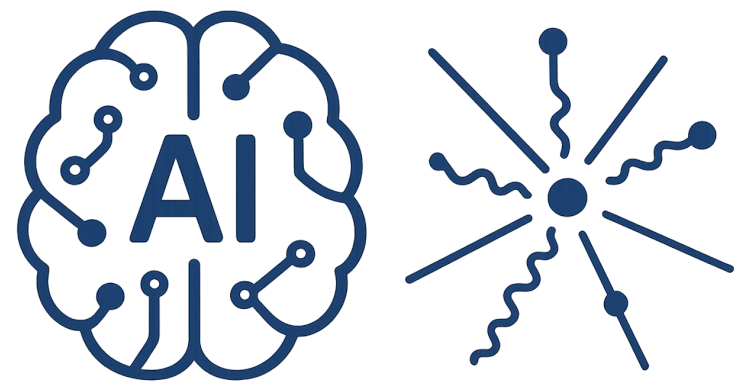
Optimization



AI-Driven HEP 6

What does an Input look like

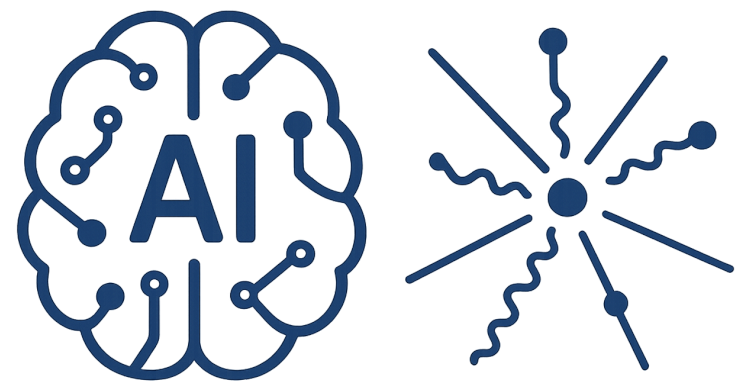
An array with shape (n, m, l, k)



AI-Driven HEP 6

What does an Input look like

An array with shape (n, m, l, k)



AI-Driven HEP 6

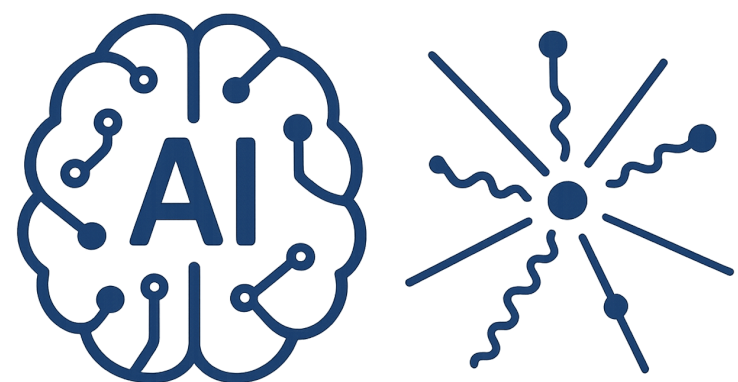
What does an Input look like

An array with shape (n, m, l, k)

(n, 1, 1, k)

n

k



AI-Driven HEP 6

What does an Input look like

An array with shape (n, m, l, k)

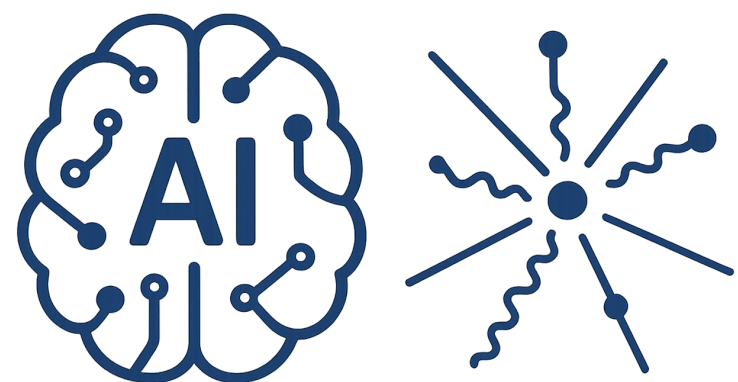
(n, 1, 1, k)

n

k

(n, 1, l, k)



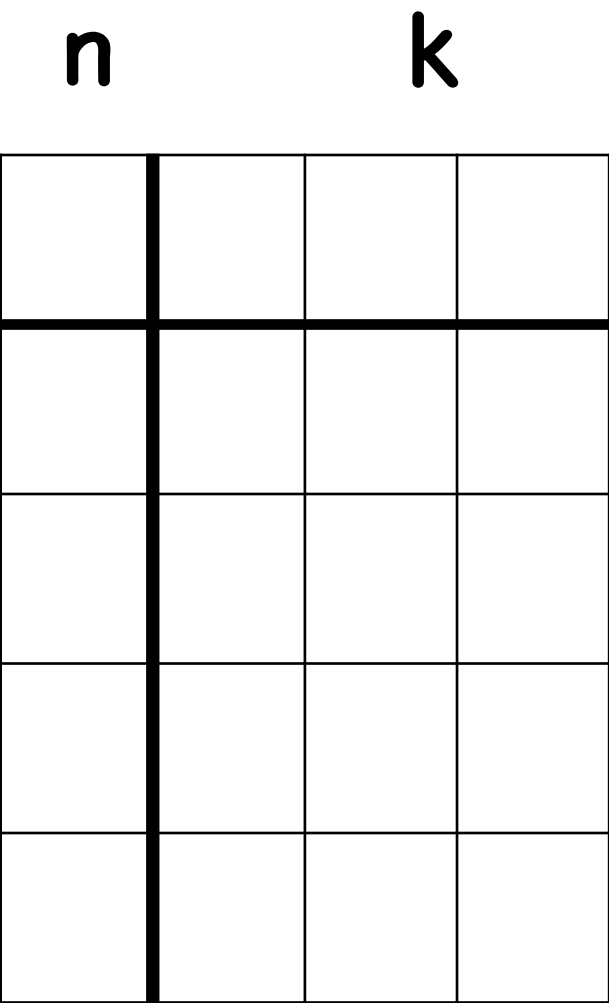


AI-Driven HEP 6

What does an Input look like

An array with shape (n, m, l, k)

(n, 1, 1, k)



(n, 1, l, k)



(n, m, l, k)

